

LIM G WEI

AI Application Developer | Backend-Oriented ML Engineer

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PROFESSIONAL SUMMARY

AI application developer focused on building practical AI-powered systems that connect backend services, databases, LLM APIs, machine learning models, and real-world workflows. Experienced in FastAPI, React, PostgreSQL, Docker, JWT authentication, workflow orchestration, Gemini API integration, CNN-based computer vision, and EEG deep learning research.

TECHNICAL SKILLS

Backend: Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, Pydantic, JWT authentication, REST APIs

AI / ML: Gemini API, LLM integration, RAG, prompt engineering, CNN, computer vision, EEG classification, model evaluation

DevOps / Testing: Docker, Docker Compose, Git, GitHub, GitHub Actions, Vercel, Render, Pytest, HTTPX, FastAPI TestClient

PROJECTS

Lifelikegame - AI-Powered Gamified Goal Management Platform | FastAPI, PostgreSQL, SQLAlchemy, Alembic, Docker, JWT, Gemini API, LangGraph, Vercel, Render

- Built and deployed a full-stack gamified life management app that turns personal goals into scheduled tasks, daily task instances, points, rewards, and AI-generated action plans.
- Designed a modular backend using routers, dependency injections, services, repositories, schemas, interfaces, and workflow orchestration for cross-module business flows.
- Implemented JWT authentication, goal and task management, daily/weekly scheduling, automated task instance generation, scoring schemes, point ledger tracking, and reward redemption.
- Integrated Gemini API and LangGraph to generate structured goal and task plans from natural language prompts with JSON-based responses.
- Frontend build from vibe coding with prompt, tech stack React

Nutrition AI - AI-Powered Nutrition Retrieval and Calorie Estimation System | PostgreSQL, pgvector, SQLAlchemy, Alembic, Docker, Gemini API

- Built a full-stack AI nutrition MVP that estimates calories from single-food text input using Gemini embeddings, pgvector similarity search, and a PostgreSQL food knowledge base.
- Implemented a retrieval-based AI workflow that embeds user input, searches nutrition records by vector similarity, returns the closest food match, and stores confidence, latency, provider, and status metadata.
- Added CSV-based food knowledge import, embedding seeding, prediction logging, user feedback, and metrics for monitoring and evaluation.
- Frontend built from vibe coding with React, while Gemini embedding, pgvector retrieval, backend workflow, and AI evaluation logic was implemented manually.

OpenLearning Course Assistant Integration - Multimodal AI Education Assistant | Python, Gemini API, Multimodal LLM, prompt engineering, logging

- Developed a private AI assistant integration for an online learning platform that allows students to submit text and image-based questions and receive text-based answers.
- Designed a multimodal prompt flow and logging structure to record group ID, prompt, AI response, and timestamp for traceability.
- Planned a course-material-first answer flow with future retrieval-augmented generation support.
- Private project; source code and platform screenshots are not publicly available due to confidentiality.

CCTV Abnormality Detection - CNN-Based Computer Vision System | Python, OpenCV, CNN, TensorFlow/Keras, image preprocessing

- Built a computer vision pipeline to identify abnormal CCTV frames using image preprocessing and CNN-based classification.
- Implemented rule-based abnormality checks using brightness and grayscale pixel ratio thresholds to support visual anomaly detection.
- Designed the system for monitoring use cases where CCTV frames are classified as normal or abnormal.
- Private project; details are summarized without exposing sensitive data.

EEG Drowsiness Detection - CNN-Based EEG Classification Research | Python, PyTorch, CNN, EEG, LOSO cross-validation, model evaluation

- Developed CNN-based models for cross-subject EEG drowsiness detection using multi-channel EEG signals.
- Evaluated model performance using Leave-One-Subject-Out cross-validation and metrics including accuracy, precision, recall, and F1-score.
- Explored channel selection and interpretability approaches for understanding EEG channel contribution in drowsiness prediction.

Work Experience

Software Engineer – Palazon Technology May 2023 – Jul 2025

- Developed and maintained the PCMS website, including bug fixes, UI adjustments, and feature improvements.
- Supported MISUMI-related product page maintenance, including product card layouts, product information updates, pricing display, and stock/status presentation.
- Worked with HTML, CSS, JavaScript, PHP / Laravel, MySQL, and phpMyAdmin to troubleshoot and maintain web application features.

RESEARCH & DEVELOPMENT FOCUS

- Backend AI systems: FastAPI services, PostgreSQL schema design, Dockerized local development, deployment configuration, and API testing.
- Machine learning applications: CNN-based visual abnormality detection and EEG-based driver drowsiness detection research.
- LLM applications: structured JSON output, multimodal question answering, prompt design, and AI planning workflows.

EDUCATION

Master of Computer Science / University Technology Malaysia / 2026

- Working on Journal Paper An Interpretable Multi-channel CNN for EEG-Based driver drowsiness detection
- Developing Open Learning AI Course Assistant
- Developing CCTV Abnormality Detection system where cooperate with Taiwan company that use ERP system on computer vision analysis

Degree / Southern University College/ 2023

- Admitted with Dean's Merit Scholarship for demonstration of excellence in Software Engineering